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Systemic/Immune-Modulation of *Olea europaea* Leaf Extract in Fetuses of Alloxan-Induced T1 Diabetic Rats

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ABSTRACT Maternal glucose is the principal macronutrient that sustains fetal growth. Prolonged exposure of the fetus to hyperglycemia from the early stages of pregnancy accelerates the maturation of the stimulus–secretion coupling mechanism in β cell autoimmunity, which leads to early hyperinsulinemia in type 1 diabetes mellitus (T1DM). Nowadays, diabetes mellitus (DM) is the most common medical complication of pregnancy, and among young women, the prevalence of overt diabetes and undiagnosed hyperglycemia is rising. Even though conventional medication is effective in treating DM, it is expensive and has harmful side effects. Herbal medicine will thus incorporate alternative therapy and be more effective and less toxic. Due to their bioactive components, olive leaves (*Olea europaea*) are frequently used medicinally; however, little is known about how this plant affects the immune system when it comes to diabetes. The current study used a pregnant mother rat model of alloxan-induced T1DM to examine the antidiabetic properties and embryonic safety of olive leaves. Forty adult female Sprague Dawley rats were split up into four groups as follows: nondiabetic, diabetic, olive, and diabetic-olive groups. All the mother rats were sacrificed on the 20th day of pregnancy, and fetuses were collected for further investigations. In diabetic pregnant mothers, fetuses had systemic modulation-negative effects. These effects were significantly reversed when the diabetic groups were supplemented with extracts from olive leaves. The findings showed that the olive leaf extract inhibits the diabetogenic effect mediated by alloxan with effective and protective systemic immunomodulation during embryonic development.

KEYWORDS: • ***Diabetic pregnant mothers*** • ***Inflammatory mediators*** • ***Insulin*** • ***Olea europaea extract***

INTRODUCTION

Chronic diabetes mellitus (DM) can result in numerous complications, injuries to different tissues, and a reduced quality of life.[1] One of the top 10 causes of death globally is DM. As a result, DM presents a special and serious risk to millions of people worldwide.[2] According to Ingelfinger and Jarcho,[3] the prevalence of diabetes is expected to rise annually to a maximum of 4.4% in 2030, with a current prevalence of 2.8% across all age groups. Pregnancy-related diabetes is linked to a higher chance of fetal, neonatal, and long-term problems for the resulting child. When a mother is diagnosed with pregestational DM, the severity of her diabetes and the timing and duration of her pregnancy-related glucose intolerance are the main predictors of the patient's outcome. Compared with gestational DM, which is defined as diabetes diagnosed during pregnancy, prediabetes has a greater negative influence on pregnancy outcomes and maternal mortality.[4,5]

Type 1 DM (T1DM) is a chronic autoimmune disease that develops as a result of complex interactions between genetic and environmental factors. Autoimmunity to β cell antigens is the first step toward the development of T1DM. Type 2 diabetes is classified into two categories. Hyperglycemia as a result of insulin insufficiency is a hallmark of T1DM. Reactive oxygen species (ROS) produced by hyperglycemia cause cell damage through a variety of pathways, and secondary complications of DM are caused by cell damage.[6,7] Maternal hyperglycemia during the first trimester and during the time of conception can result in diabetic embryopathy, which can cause significant birth defects and

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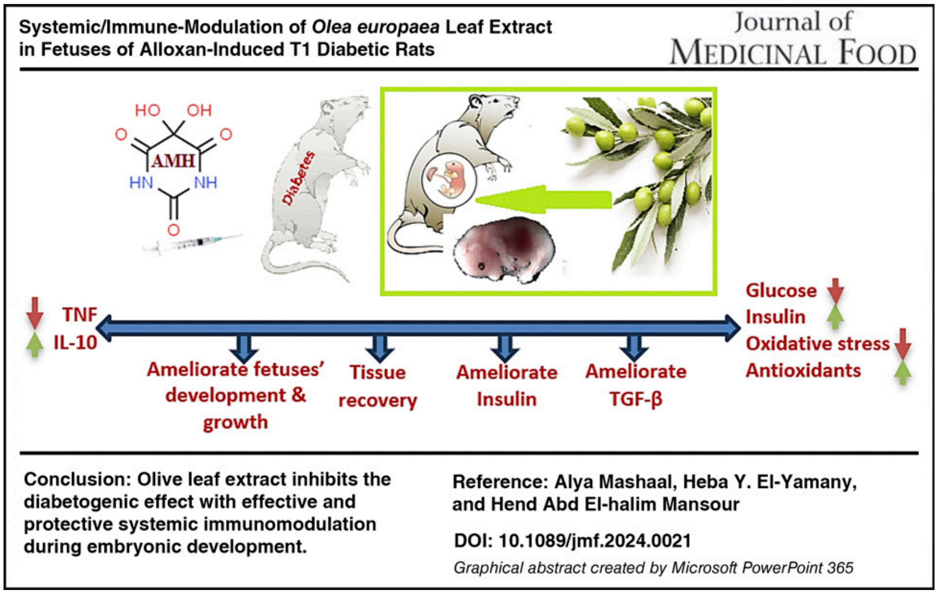

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increase the risk of spontaneous abortions and congenital malformations. Furthermore, the fetus is primarily exposed to hyperglycemia, which slows down fetal growth in the absence of secondary hyperinsulinemia. Maternal body mass index and fasting blood glucose levels both rise with the risk of malformations. This mostly happens in pregnancies where pregestational diabetes is present.[8] Fetal hyperglycemia, hyperinsulinemia, and macrosomia are the outcomes of diabetic fetopathy, which develops in the second and third trimesters.[9] In the second trimester, hypertrophic islet cells produce more insulin in response to hyperglycemia. A powerful cascade of third-trimester events culminating in a notable increase in fat and protein stores is caused by the potent combination of hyperglycemia, a major anabolic fuel, and hyperinsulinemia, a major anabolic hormone.[10,11] According to Calkins and Devaskar,[12] a hyperglycemic mother's environment has also been linked to fetal growth retardation and an increased risk of negative health outcomes for the newborn, including obesity, insulin resistance, type 2 DM, and metabolic syndrome. Pregestational T1DM has been associated with mild neurological and cognitive deficits in offspring, such as increased symptoms of attention deficit hyperactivity disorder, impaired fine and gross motor skills, and impaired explicit memory performance.[13,14]

The use of alloxan and streptozotocin (STZ) in inducing T1DM models[15,16] serves different purposes and comes with distinct advantages and limitations. In choosing between alloxan and STZ models, many factors should be considered such as the desired onset and duration of diabetes, the extent of beta cell destruction, and the specific research questions being addressed. Although both models are valuable for studying T1DM pathophysiology and evaluating potential therapeutic interventions, the specific requirements of the study design and objectives is the major key player to choose between alloxan or STZ.

However, STZ is commonly used to induce T1DM models due to its ability to specifically damage pancreatic beta cells through DNA alkylation[17] and allows for the study of autoimmune mechanisms involved in β cell destruction.[18] Alloxan selectively targets and destroys pancreatic beta cells, leading to insulin deficiency and hyperglycemia, mimicking the pathophysiology of T1DM. One of the advantages of alloxan is its rapid onset of action, which allows for quick induction of diabetes and cost-effectiveness. In addition, alloxan-induced diabetes closely resembles the autoimmune destruction of beta cells seen in human T1DM.[15]

T1DM is induced by alloxan (alloxan, β cell-specific cytotoxins) in animal models.[15] Alloxan has a unique action to suppress glucokinase activity and generate ROS that significantly has necrotizing effect on β cells.[18] Despite recent improvements in the management of diabetes, the use of currently available medications is still linked to a number of complications, such as hypoglycemia, cardiovascular risks, increased risk of morbidity, and gastrointestinal disorders.[19,20]

Plants contain a wealth of bioactive substances; they make attractive candidates for managing DM. In recent years, a great deal of research has been done to find or isolate natural substances that are less likely than synthetic medications to cause negative side effects while still being effective in managing diabetes.[21] Potential natural sources of novel hypoglycemic compounds could be found among the antidiabetic plants. Because these plants are safe and have been shown to help treat intrauterine growth retardation, they are consumed in greater quantities during pregnancy.[22] It is true that medicinal plants are economically viable, exhibit excellent biological activity, and have low toxicity.[23]

Olive leaves have gained attention in the management of T1DM due to their potential health benefits. Some studies suggest that olive leaf extract may help in controlling blood sugar levels by improving insulin sensitivity and reducing

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inflammation.[24–26] The active compound in olive leaves, oleuropein, has been studied for its antioxidant properties, which may help protect pancreatic beta cells and improve insulin secretion.[10]

However, it is essential to note that research on the specific use of olive leaves in T1DM is still in its early stages, and more studies are needed to confirm its effectiveness and safety in managing this health problem. Thus, this study aimed to evaluate the potential antidiabetic properties and embryonic safety of olive leaf extract in a pregnant rat model of alloxan-induced T1DM and also highlight the potential of olive leaf extract to inhibit the diabetogenic effects and reverse the systemic immunomodulation during embryonic development.

MATERIALS AND METHODS

Ethical approval

The National Hepatology and Tropical Medicine Research Institute, Cairo, Egypt, Research Ethics Committee approved the protocol for the experiment, which was carried out in accordance with the ARRIVE guidelines 2.0 (serial No. A3-2023). The National Institutes of Health's Guide for the Care and Use of Laboratory Animals (NIH Publication No. 85-23, revised 1996) was followed in every step of the process.

Experimental animal feeding and maintenance

Forty adult female Sprague Dawley rats (weighing between 160 ± 20 g) were acquired at the age of 8 ± 1 week from the Egyptian Holding Company for Biological Products and Vaccines (located in Cairo-Helwan, Egypt) animal house. For 2 weeks prior to the experiment, the rats were housed in an environment with the right amount of light, temperature, and ventilation. The rats were fed a typical diet and had unlimited access to food and water.

Alloxan diabetes induction

Single subcutaneous injection of 120 mg/kg body weight of alloxan monohydrate (AMH, El-Gomhouria Company, Cairo, Egypt) in ice-cold citrate buffer (0.01 M; pH 4.5) was administered to female rats to induce T1DM according to Mostafavinia et al.[27] Using a glucometer (LifeScan Inc., Milpitas, CA, USA), blood glucose levels in the animals were tracked for the following seven days following injection. As controls, an equal number of animals with similar weights were kept. The animals were confirmed to have diabetes, as defined by fasting blood glucose levels greater than 250 mg/dL.[28]

Estrus cycle and mating

The female had gone through at least two consecutive estrus cycles prior to use, and the vaginal smear technique was used to determine her estrus cycle stage. The study excluded the females who failed to mate within two estrus cycles. Male rats without diabetes and female rats with diabetes were bred together 1:1. Every female was examined for vaginal plugs. When the first day vaginal plug was visible, it was considered gestational day 0. On the 20th day of pregnancy, all pregnant rats were dissected, and the fetuses were collected for further study.[29]

Olive leaves extract

The powdered olive leaves were purchased from El-Gomhouria Company located in Cairo, Egypt. The phenolic compounds present in olive leaves were identified using high-performance liquid chromatography (HPLC) analysis, 3000 HPLC system; Thermo Fisher Scientific, following the methodology outlined by Biswas et al.[30]

The dose of olive leaf extract (OLE) used in this study is carefully selected to optimize therapeutic efficacy, ensure consistency in bioactive compound content, and minimize the risk of adverse effects, thereby providing a balance between efficacy and safety, keeping in mind the pregnancy of the rats. When determining the dose of OLE for pregnant diabetic rats, several factors have been considered, particularly the safety of both the mother and the developing embryos. The dose is carefully selected to ensure that it effectively modulates diabetes-related parameters without exerting harmful effects on pregnancy outcomes. In addition, previous studies examining the use of OLE in diabetic animal models should be taken into account. Hassan et al.[31] provide valuable insights into appropriate dosing regimens; an oral stomach tube was used to administer a freshly prepared aqueous solution containing 55 mg/mL of distilled water at a dose of 150 mg/kg of body weight.

The human equivalent dose, related to the body surface area normalization method based on FDA draft guidelines, is 24.3 mg/kg of body weight.[32]

Experimental design

The rats were randomly and equally distributed into two main categories as follows: the nondiabetic and diabetic pregnant rats, each of them further divided into two groups of 10 rats per group. The nondiabetic pregnant rats were subdivided into control group injected with ice-cold citrate buffer (0.01 M; pH 4.5) (C), and group 2, rats received OLE orally (150 mg/kg of b.wt.) for 15 days, (OLE). Diabetic pregnant rats were subdivided into group 3, rats injected with alloxan (AMH), and group 4, rats injected with alloxan then administrated OLE orally (150 mg/kg of b.wt.) for 15 days (AMH-OLE). The schedule was repeated from the 5th day of pregnancy (the day of pregnancy confirmation) until the 20th gestational day, where the dams were sacrificed, and their embryos were separated from wombs for further studies.

Blood collection and tissue sampling

Blood was taken from the retro-orbital plexus of dams immediately following the anesthesia of all groups' animals with 1.5% isoflurane.[33] After coagulating at room temperature for 10 minutes, blood samples were centrifuged (Hermle Z206-A Compact Centrifuge, US) for 10 minutes at 3000 rpm to produce a clear, nonhemolyzed serum for biochemical analysis. The fetus was promptly dissected, and tissues were ready for additional histological and immunohistochemical analyses and investigations.

Demographical studies

Number of implantation sites
Number of resorbed bodies
The anal nasal tall (fetal length)

Biochemical and oxidative analyses

Serum samples of dams and embryo tissue homogenates were used to evaluate activities of glucose (DetectX® Glucose Colorimetric Detection Kit, Thermo Fisher Scientific Inc., Cat. # EIAGLUC, Invitrogen, Carlsbad, CA, USA) and insulin (Rat Insulin ELISA Quantification Assay Kit, Thermo Fisher Scientific Inc., Cat. # ERINS, Invitrogen, Waltham, MA, USA). Glycogen level was detected in dams' livers and embryo tissue homogenates (Glycogen Quantification Assay Kit, Cat. # MAK016, Sigma-Aldrich, Saint Louis, MO, USA). In addition, oxidative activities of superoxide dismutase (SOD Colorimetric Assay Kit, Cat. # ab65354, Abcam, Cambridge, UK), glutathione peroxidase (GPx Colorimetric Assay Kit, Abcam, Cat. # ab102530, Cambridge, UK), catalase (Colorimetric Activity Kit, Thermo Fisher Scientific Inc., Cat. # EIACATC, Invitrogen, Fredrich, MD, USA), and malondialdehyde (MDA Colorimetric Assay Kit, Cat. # ab118970, Abcam, Cambridge, UK).

Inflammation assays

Sandwich enzyme-linked immunosorbent assay was used to detect inflammatory mediators' cytokine, tumor necrosis factor-α (TNF-α, Thermo Fisher Scientific Inc., Cat. # KRC3011, Invitrogen, Vienna, Austria), and interleukin-10 (IL-10, Thermo Fisher Scientific Inc., Cat. # ERA23RB, Invitrogen, Carlsbad, CA, USA) in embryo tissue homogenates. The optical density was read on a microtiter plate reader with 450 nm (ELX-808, BioTek Instruments, Winooski, VT, USA).

Histopathological and immunohistochemical analyses

The diabetic and nondiabetic group mothers were sacrificed on the 20th day of gestation. Fetuses from various groups were obtained, fixed in Carnoy's fluid and neutral buffer formol, and then dehydrated in upgrading concentration of alcohol, cleared in xylene, and waxed in paraffin. The fixative was injected to cover the entire organ. Hematoxylin and eosin, a general histological stain, was used to examine liver tissues in 5-μm sections, Mallory's trichrome stain was used to identify collagen fibers,[34] and Periodic acid Schiff's (PAS) technique, PAS positive materials was used to detect polysaccharides such as glycogen.[35] Immunohistochemical methods were applied to detect insulin marker (Insulin Polyclonal antibody, Thermo Fisher Scientific Inc., Cat. # PA5-120784, Invitrogen), transforming growth factor-β (TGF-β; anti-TGF-β, Thermo Fisher Scientific Inc., Cat. # MA5-16949, Invitrogen), and apoptotic activities with caspase-3 (anti-caspase 3 antibodies, Thermo Fisher Scientific Inc., Cat. # 43–7800, Invitrogen).

Statistical analysis

The statistical analysis for social science (version 23) program, SPSS, was used to process and analyze the data. The difference between means was evaluated by one-way analysis of variance followed by *post hoc* test, least significance difference (LSD). The means ± standard deviations of the data are displayed, and statistical significance was determined by $P \le .05$.

RESULTS

The main phenolic compounds are identified in the olive leaves extract; the specific analytical results obtained from HPLC are typically measured in mg/100 g of the extract. Commonly found phenolic compounds in olive leaf extracts include rutin, 248.22; quercetin, 130; oleuropein, 115.33; catechin, 55; apigenin, 50; luteolin, 40.54; caffeic acid, 35.33; hydroxytyrosol, 30.96; and tyrosol, 13.26.

Demographical evidence

Number of implantation sites. Number of implantation sites' mean values on the 20th day of gestation are represented in Figure 1A that showed a nonsignificant decrease ($P \le .05$) in pregnant rats of OLE group, highly significant decrease ($P \le .001$) in AMH group, and significant decrease in AMH-OLE ($P \le .05$) compared with the control mothers (C), which reached 8.5 ± 1.38, accompanied with decrease in the percent of change −2.3% in the OLE group, −40% in the AMH group, and −29.4% in the AMH-OLE group.

Number of resorbed bodies. Figure 1B represents the current results of the mean value of resorbed bodies that showed a nonsignificant increase ($P \le .05$) in OLE group, highly significant increase ($P \le .001$) in the AMH group, and a significant increase ($P \le .05$) in AMH-OLE group compared with the control group with increased percent of change 66.6% in the OLE group, 666% in the AMH group, and 266% in the AMH-OLE group.

The anal nasal tall (fetal length). Anal nasal tall (fetal length) mean values of the present study are represented in Figure 1C. The results showed no significant changes ($P \le .05$) in the OLE group and a highly significant decrease in the AMH and AMH-OLE groups compared with the healthy control (5.9 cm). The percent of change of fetal length was −1.69% in group OLE group and increased to −35.5% in the AMH group, whereas the AMH-OLE group percent decreased to −12.5%.

Biochemical and oxidative compounds

The serum analysis of dams (Fig. 2A) showed the present results as very highly significant increases ($P \le .001$) in the mean glucose levels of the AMH and AMH-OLE groups compared with the healthy control group. The mean insulin level showed very highly significant decreases in the AMH group, whereas the AMH-OLE group showed slight reduction compared with the control group. The mean liver glycogen

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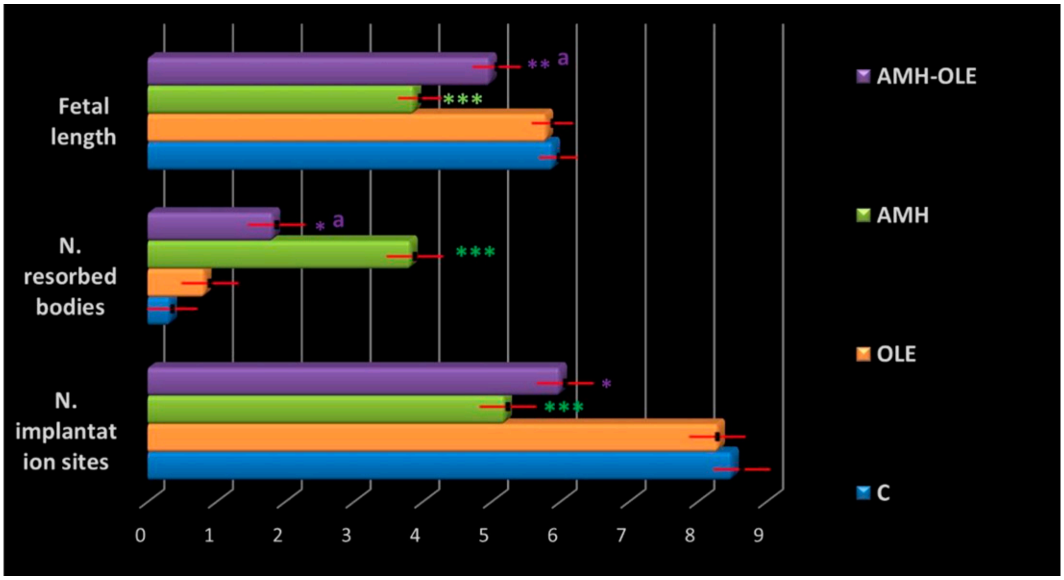

FIG. 1. Average values diagrammatic representation of the number of implantation sites, the number of resorbed bodies, and the fetal length in control and treated groups. Each value represented the mean and ± standard deviation (SD), $n = 6$ rats. * Significant difference vs. the control group at $P < .05$, ** highly significant difference vs. the control group at $P < .01$, *** very significant difference from the mean of the control group at $P < .001$. [a]Significant difference vs. the diabetic group at $P < .05$. C, control group; OLE, olive leaf extract only administrated group; AMH, diabetic group; AMH-OLE, diabetic group administrated OLE.

levels of the diabetic groups showed very highly significant decreases, in comparison with the control group. OLE supplementation slightly reduces glucose and increases insulin levels that mitigate the decrease in liver glycogen levels observed in diabetic groups compared with healthy controls. In addition, oxidative biomarkers showed very highly significant decreases in SOD levels of both diabetic groups accompanied with concentration recovery in the AMH-OLE group. GPx mean levels showed very highly significant decreases in the diabetic group, whereas the AMH-OLE group showed significant reduction compared with the control group. AMH group showed significantly decreased CAT levels compared with control group, whereas the AMH-OLE group almost recovered to reach normal value. A very highly significant increase in MDA mean levels of the diabetic group was noticed, whereas the AMH-OLE group recovered to show significant reduction in comparison with the control group. OLE supplementation demonstrates rescuing the effects of AMH on oxidative biomarkers. This is evidenced by the recovery in antioxidant markers and restoration of oxidant biomarker levels to near-normal values.

The present results of embryos supernatant analysis (Fig. 2B) showed very highly significant increases ($P \leq .001$) in the mean glucose levels of the AMH groups, whereas the AMH-OLE group showed significant elevation ($P \leq .05$) compared with the healthy control group. The insulin level showed significant reduction ($P \leq .05$) in the diabetic group compared with the control group, whereas the AMH-OLE group almost recovered to reach normal value. The glycogen concentrations showed a significant reduction in the AMH group in comparison with the control group. OLE mitigates the effects of AMH on embryos' supernatant parameters. It shows potentially counteracting hyperglycemia-induced effects and maintains nearly normal insulin levels and partially preserves glycogen concentrations, suggesting its protective role against diabetes-induced alterations. Oxidative amelioration of olive leaf administration of the diabetic group showed highly significant decrease ($P \leq .01$) in SOD level and very highly significant decrease ($P \leq .001$) in GPx and CAT levels compared with the healthy control in contrast to MDA level, which showed significant elevation. The oxidative results of this group exhibited a concentration recovery compared with the AMH group, which showed extremely significant decrease in all of SOD, GPx, and CAT mean levels, as opposed to extremely significant increase in MDA level compared with the findings in the control group. OLE counteracts the oxidative effects of AMH in embryos' supernatant, by restoring antioxidant enzyme levels, and reduces oxidative biomarker levels, indicating its potential to alleviate oxidative stress in diabetic embryos.

Olive leaf administration alone exerts beneficial effects both in dams and embryos; OLE group has been found to have antioxidant properties by reducing MDA level as oxidative marker and increasing the expression of SOD, GPx, and CAT levels

FIG. 2. Glucose/insulin sensitivity and oxidative stress mediators in control and treated groups. **(A)** circulating mean values of mediators in mothers. **(B)** Tissue mean values of mediators in embryos. Each value represented the mean ± standard deviation (SD), $n = 6$ rats. * Significant difference vs. the control group at $P < .05$, ** highly significant difference vs. the control group at $P < .01$, *** very significant difference from the mean of the control group at $P < .001$, [a] Significant difference vs. the diabetic group at $P < .05$. SOD, superoxide dismutase; GPx, glutathione peroxidase; CAT, catalase; MDA, malondialdehyde; C, control group; OLE, olive leaf extract only administrated group; AMH, diabetic group; AMH-OLE, diabetic group administrated OLE.

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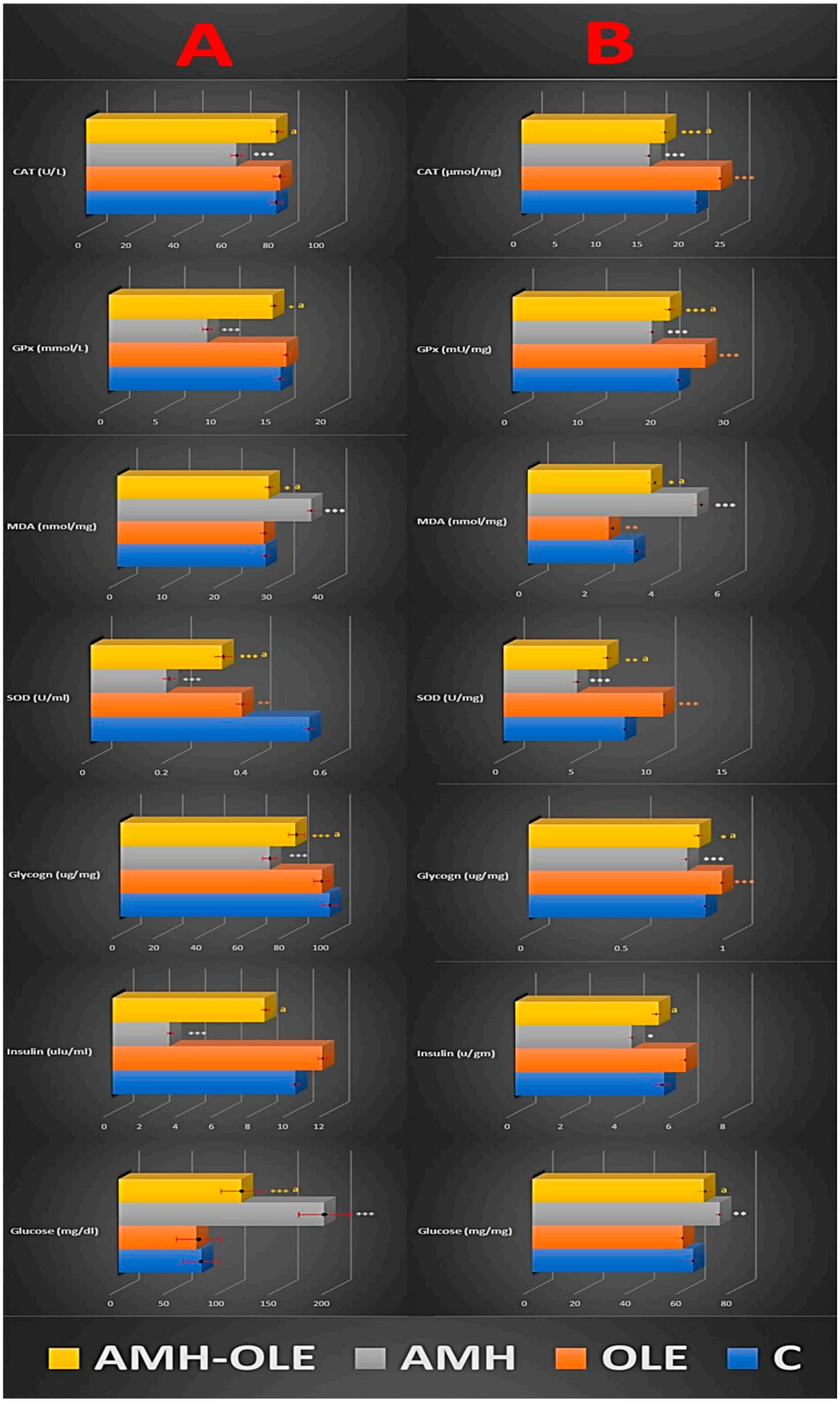

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in the current results. Further regulation of glucose levels was seen by improving insulin sensitivity and increasing glycogen storage (Fig. 2A&B).

Inflammatory agents

Figure 3 illustrates the impact of consuming OLE on the concentrations of TNF-α and IL-10 cytokines, as markers of inflammatory process in diabetic/nondiabetic mother rats. TNF-α concentration in the AMH group showed very highly significant increase, whereas the TNF-α concentration in the AMH-OLE group was significantly increased compared with the diabetic group. The IL-10 expression compared among the control, AMH, and AMH-OLE diabetic groups were significantly decreased (Fig. 3).

OLE results show potential in modulating inflammatory markers in diabetic mother rats induced by AMH, by significantly reducing pro-inflammatory response and partially mitigating the decrease in anti-inflammatory expression observed in diabetic groups.

Histopathological and immunohistochemical

Fetal hepatic tissues of the control and olive groups showed normal structure of the central vein, cords of hepatocytes, sinusoidal spaces (Fig. 4a & b) with normal distribution of thin bundles of collagen fibers supporting the hepatocytes, sinusoidal spaces, and wall of the central vein (Fig. 4e & f), whereas polysaccharides recorded normal staining affinity (Fig. 4i & j).

Fetal liver tissue of diabetic group showed numerous necrotic areas with degenerated hepatocytes, lots of nucleated red blood cells (RBCs; micronuclei), highly dilated and ruptured wall of the hepatic portal vein, and blood sinusoids, which contained hemolyzed blood cells and numerous pyknotic nuclei (Fig. 4c). In addition, decreased collagen fibers (Fig. 4k) and increased polysaccharides (Fig. 4g) were realized within hepatic tissues.

The fetal liver tissue of diabetic group treated with OLE exhibited tissue recovery with somewhat normal appearance (Fig. 4d), increased collagen fibers especially in the wall of the central vein and the capsule (Fig. 4h), and normal polysaccharide content within the tissue (Fig. 4l).

Immunohistochemical examination of fetal livers for insulin detection of control and OLE groups revealed active positive cytoplasmic stain ability of hepatocytes $^{+3}$ve and $^{+4}$ve, respectively (Fig. 5a, b). The diabetic rat's fetal livers showed less immunoreactivity $^{+1}$ve with 10–35% high-power field (HPF) positively reacted cells with insulin diabetic marker and the remaining cells 65–90% HPF negatively stained (Fig. 5c). Meanwhile, the AMH-OLE group revealed positively immunoreacted hepatocytes $^{+2}$ve compared with the AMH group as 20–45% HPF positively stained hepatocytes and 55–80% HPF negatively stained (Fig. 5d).

Most of the hepatocytes of fetal livers of control and OLE groups were normally expressed positive stained TGF-β ($^{+1}$ve) (Fig. 5e, f). Sections from fetal livers of diabetic mothers showed highly expressed immune reactivity against TGF-β ($^{+4}$ve; 60–70% HPF) (Fig. 5g), whereas the OLE-diabetic treated group showed partial enhancement of TGF-β immune expression ($^{+3}$ve; 45–65% HPF) (Fig. 5h).

DISCUSSION

Maternal hyperglycemia and impaired carbohydrate metabolism are hallmarks of DM. In the current investigation, we calculated OLE's potential as an antidiabetic by comparing it to the fetotoxicity that diabetes causes in rat fetuses.

The study consistent with Mohammed et al.[11] revealed that the implantation sites in the OLE group were similar to the control group, whereas the AMH and AMH-OLE groups showed decreased implantation sites and increased resorbed bodies. Insulin resistance due to gestational diabetes can lead to increased glucose transfer to the fetus, potentially causing fetal hypoinsulinemia, hyperglycemia, and increased risks of

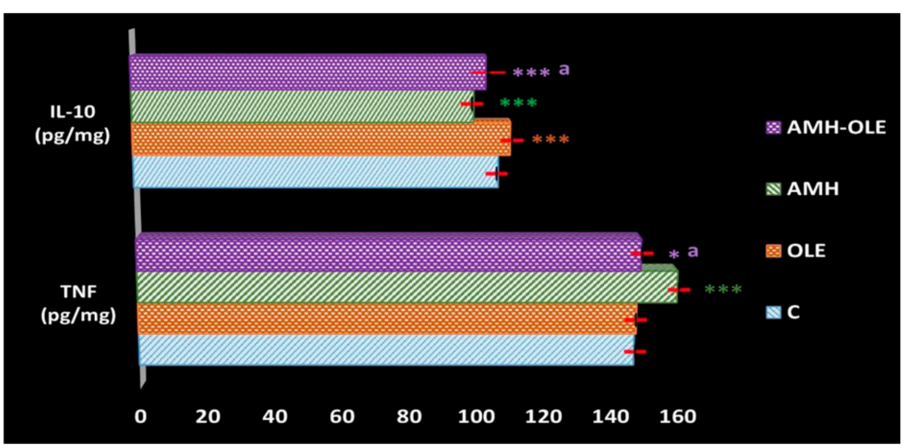

FIG. 3. Inflammatory hemostasis expression in embryos tissue of control and treated groups. Each value is presented as the mean ± standard deviation (SD), n = 6 embryos. * significant difference vs. the control group at P < .05. *** very significant difference vs. the control group at P < .001, [a]Significant difference vs. the diabetic group at P < .05. TNF-α, tumor necrosis factor-alpha; IL-10, interleukin-10; C, control group; OLE, olive leaf extract only administrated group; AMH, diabetic group; AMH-OLE, diabetic group administrated OLE.

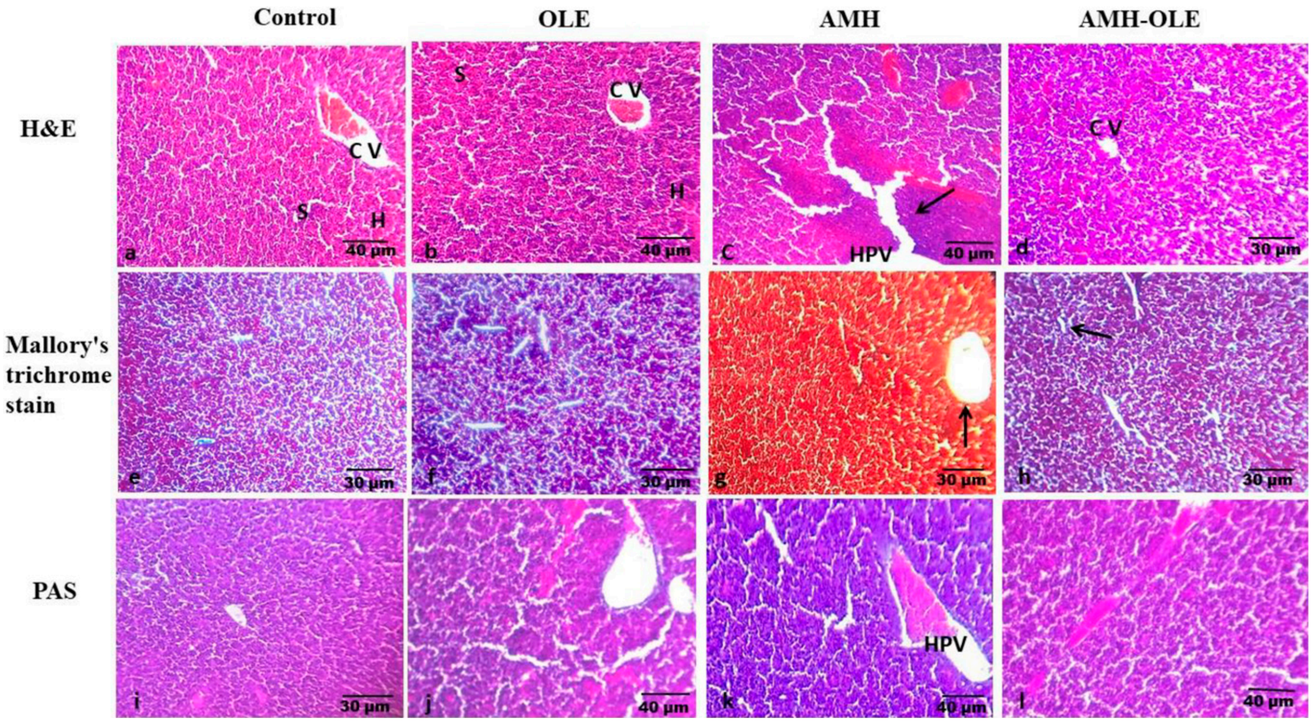

FIG. 4. Histopathological and histochemical investigations. **(a, b)** Fetal liver tissues of the control and olive groups, respectively, showing normal architecture of the central vein (cv), cords of hepatocytes **(H)**, and sinusoidal spaces **(s)**. **(c)** Fetal liver tissue of AMH group with highly dilated and ruptured wall of the hepatic portal vein and blood sinusoids, which contain hemolyzed blood cells, numerous necrotic areas which contain degenerated hepatocytes, lots of karyolitic cells (*arrow*). **(d)** Normal appearance of hepatocytes, endothelial lining of the central vein of fetal liver tissue of AMH+OLE group. **(e, f)** Fetal liver tissues of the control and OLE groups, respectively, showing normal distribution of collagen fibers supporting the hepatocytes and sinusoidal spaces. **(g)** Highly decreased collagen fibers in the fetal liver tissue of AMH group, especially around the wall (*arrow*) of the hepatic portal vein. **(h)** Increased collagen fibers in the liver tissue of AMH+OLE group, especially in wall of the central vein (*arrow*) and the capsule. **(i, j)** Normal distribution of PAS +ve materials in the fetal liver tissues of the control and olive groups. **(k)** Increased staining affinity of PAS +ve materials in hepatocytes with negatively stained degenerated areas of diabetic group. **(l)** Moderately stained PAS +ve materials in the fetal liver tissue of AMH+OLE group.

fetal death and growth restriction.[36] Pregnant women with pregestational T1DM face higher risks of miscarriage, stillbirth, congenital defects, placental anomalies, intrauterine malprogramming, and fetal impairments, as supported by the study by Jawerbaum and White,[37] which highlighted reduced placental blood flow during hypertensive disorder of pregnancy leading to fetal growth restriction and increased fetal mortality.[38]

The current study found that fetal length was shorter in the AMH group compared with the healthy control group, but there was no significant difference in the OLE or AMH-OLE groups; this finding is aligned with Mohammed et al.[11] Regarding the impact of maternal hyperglycemia on fetal body size, Rudge et al.[39] noted that maternal diabetes creates an unfavorable environment for embryonic development. These findings suggest that OLE may play a crucial role in regulating glucose hemostasis and promoting fetal development, as indicated by previous studies.[40,41]

This study indicated that OLE improves blood sugar control by enhancing insulin sensitivity and glycogen storage in those worst affected by pregestational diabetes. OLE activates hepatic AMP-activated protein kinase signaling, repairs islet morphology, modulates insulin secretion, and enhances glucose tolerance.[24] In infants born to mothers with pregestational DM, high maternal blood glucose levels can stimulate insulin production, affecting glucose regulation through various mechanisms such as glucose uptake modulation, inhibition of glucosidase and amylase, gene expression alteration, hormone action stimulation, and reduction of oxidative stress.[42,43] Severe hyperglycemia in diabetic rats may result from hypoinsulinemia due to cytotoxicity toward β cells, causing β cell vacuolation.[20] T1DM inhibits glucokinase activity and induces insulin-dependent diabetes by stimulating ROS production through a cyclic reaction with dialuric acid.[44] Islet cells exposed to diabetic induction show increased lipid peroxidation, reduced cell viability, elevated mitochondrial membrane potentials, and higher levels of peroxynitrite, nitric oxide, and ROS.[44]

Antioxidant qualities of OLE were discovered in the current study, which may aid in lowering oxidative stress. Accordingly, oleuropein and hydroxytyrosol, which are antioxidants found in OLE, are effective in preventing certain metabolic disorders linked to oxidative stress, such as diabetes.[10] El-Nabarawy[45] also found that the water extract from olive leaves reduced DNA damage and oxidative stress in the embryos and placentas of the diabetic mothers. OLE has antioxidant properties, particularly through

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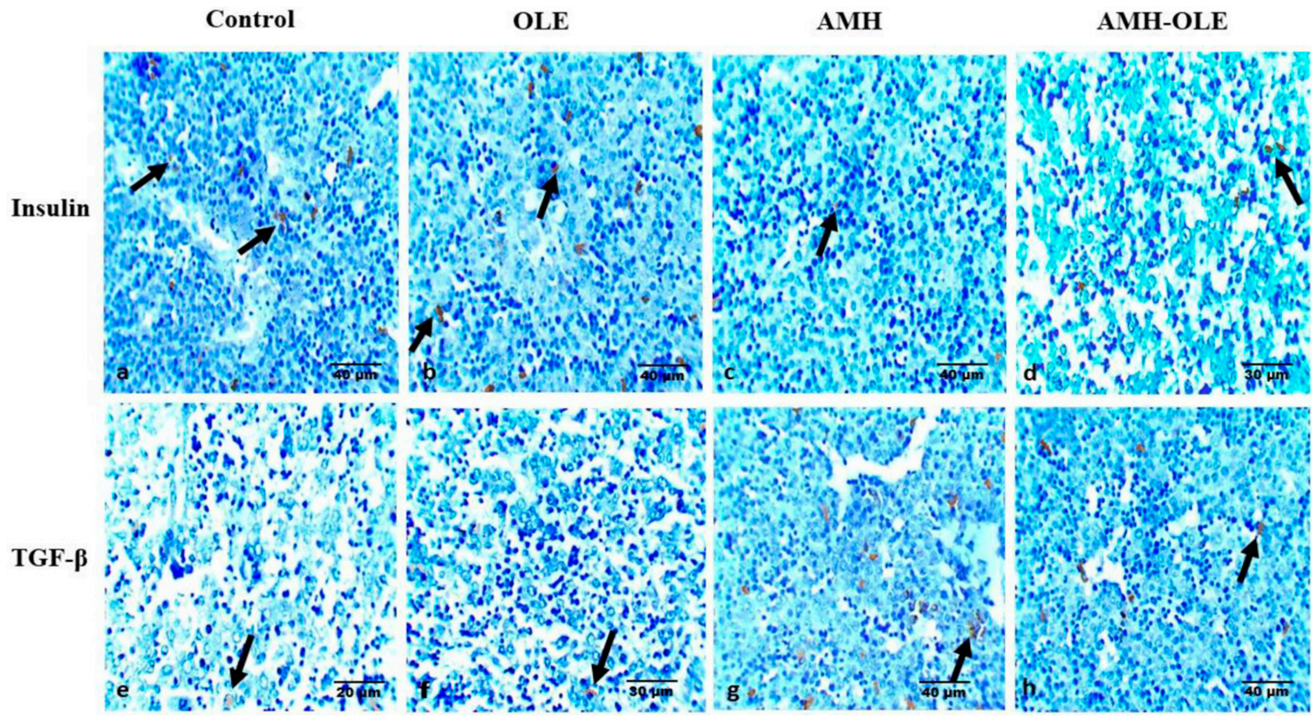

FIG. 5. Immunohistochemical examination of fetal livers for insulin and TGF-β. **(a, b)** Control and OLE groups, respectively, revealed active positive cytoplasmic stain ability of hepatocytes (*black arrow*). **(c)** The diabetic rat's fetal livers showed less immunoreactivity with positively reacted cells with insulin (*black arrow*). **(d)** AMH-OLE group revealed positively immunoreacted hepatocytes (*black arrow*). **(e, f)** Most of the hepatocytes of fetal livers of control and OLE groups, respectively, were normally expressed positive stained TGF-β. **(g)** Sections from fetal livers of diabetic mothers showed highly expressed immune reactivity against TGF-β (*black arrow*). **(h)** OLE-diabetic treated group showed partial enhancement of TGF-β immune reactivity.

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compounds such as oleuropein and hydroxytyrosol, which can lower oxidative stress and prevent metabolic disorders such as diabetes.[10]

Current findings indicate that olive leaves possess anti-inflammatory effects that are supported by studies of Asghari et al.[25] and Hadrich et al.[26] These effects are important for exploring the safety of olive leaf extracts in pregnant animals, potentially aiding in preventing inflammatory diseases and pregnancy-related complications.[46] Pedraza-Chaverri et al.[47] noted that oxidative stress and inflammation play key roles in various diseases, highlighting the importance of agents that can control these processes. The balance between TNF-α and IL-10 is crucial in maintaining inflammation homeostasis, as their imbalance can lead to a feedback loop of inflammation and oxidative stress.[48,49]

OLEs directly inhibit the pro-inflammatory cytokine TNF-α by suppressing its mRNA expression through the nuclear factor-kappa beta (NF-κB) pathway. This inhibition is facilitated by the G-protein-coupled bile acid receptor, Takeda G protein-coupled receptor 5 (TGR5), which regulates various metabolic and signaling pathways related to inflammation.[50,51] Oleanolic acid, a key component of olive leaf extracts, acts as a TGR5 agonist, activating the JNK pathway and controlling inflammatory cytokine production.[52,53] This mechanism, along with oleanolic acid's potent anti-inflammatory effects, contributes to the suppression of inflammation and tissue damage. Oleuropein improved glucose tolerance, decreased insulin levels, reduced inflammation, and enhanced insulin signaling in insulin-resistant rats.[26] OLEs also show promise in reducing oxidative stress and inhibiting inflammation.[54,55]

The current findings show normal architecture and normal collagen fiber distribution of fetal liver tissue for both control and OLE groups that align with findings of Khalid et al.[56] and Fornes et al.[57] In this study, the diabetic group showed fetal liver tissue pathogenicity, possibly due to alloxan-induced cellular damage. Alloxan's cytotoxic effects involve free radical production, DNA alkylation, and subsequent fetal hypoxemia.[57] Treatment with OLE in diabetic mothers improved histological examination by reducing lipid metabolic enzymes and peroxisome proliferator-activated receptor (PPAR) coactivators, suggesting benefits for fetal liver development. OLEs potentially modulate oxidative and inflammatory mediators that improve apoptosis in multiple organs.[54,55]

The current study demonstrated increased staining affinity of PAS $^{+}$ve materials in hepatocytes of fetal liver tissues of diabetic mothers, which is a method used to detect polysaccharides such as glycogen and mucosubstances such as glycoproteins, glycolipids, and mucins in tissues. It is evident that disorders associated with diabetes cause impairments to all of the liver's metabolic functions in fetuses. When the liver cannot store glycogen and cells are unable to use glucose, hyperglycemia results. Moreover, the production or release of ROS and the reduction of antioxidant reserves in

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DM increased oxidative stress. Diabetes-related liver disease is known as diabetic hepatopathy. The weight of the liver increases as diabetes worsens because of fibrosis or abnormal glycosylation linked to hepatic alloxanatosis.[58] However, the current findings in the AMH-OLE group showed an improvement in the amount of carbohydrates compared with the group with diabetes. This could be the result of enhanced liver functions, which supply extrahepatic tissues and mediate the conversion of food energy.[59–61]

In the current study, fetal liver immunohistochemical analysis showed positive insulin expression in the control and OLE groups, decreased expression in diabetic rats, and partial improvement in the OLE-diabetic group. This aligns with Tangvarasittichai's observations linking hyperglycemia associated with insulin deficiency to biochemical processes such as oxidative stress, inflammation, and apoptosis. Insulin immunoreactivity is significantly reduced in animal models of diabetes, but olive extract consumption increases insulin sensitivity, likely due to its antioxidant and anti-inflammatory properties.[62]

The current study revealed normal expression of TGF-β in control and OLE fetal liver tissue. Oleanolic acid positively affects TGF-β expression, crucial for regulating cell growth and differentiation during fetal development. TGF-β plays a role in developing functional liver tissue by maintaining tissue homeostasis, although its expression and roles can vary with context and developmental stage.[63,64] The increased expression of TGF-β in diabetic fetal liver hepatocytes is primarily due to diabetes-induced molecular reactions, including elevated blood glucose levels activating the TGF-β/Smad signaling pathway. This pathway is further intensified by chronic inflammation and oxidative stress associated with diabetes, as TGF-β regulates both processes. In addition, TGF-β's role in tissue repair and regeneration leads to its overexpression as a feedback mechanism to prevent tissue damage and promote healing in diabetic hepatocytes.[65,66]

Consuming olive leaves, particularly compounds such as oleuropein and hydroxytyrosol, can regulate TGF-β expression through their anti-inflammatory and antioxidant properties. These substances may reduce oxidative stress and inflammation, reducing the need for TGF-β in tissue repair. Olive leaf components can also directly influence TGF-β signaling pathways, maintaining balanced expression levels by mitigating inflammation and oxidative stress. Furthermore, oleuropein has been shown to reverse cell apoptosis, regenerate tissue, reorganize histological structure, and lower oxidative stress in diabetes complications.[67,68]

CONCLUSION

Olea europaea extract modulates immune and systemic pathways and has a protective effect on the embryos of T1DM mothers. Its active compounds play a pharmacological role in embryonic development and diabetic recovery. This suggests that OLE could serve as a protective agent against hyperglycemia damage and inflammation in diabetic pregnancies. Further research into gene expression and inflammatory signaling molecules is needed to understand its mechanism of action and impact on embryonic development. In addition, a dose–response assessment for OLE in pregnant diabetic animal models would be prudent to conduct for better understanding of therapeutic range and potential side effects.

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